



Some of the participating teams and their rovers

and climbed into the craters of active volcanoes in Alaska and Antarctica.

Polaris and Red Rover are the two rovers being developed by Astrobotic.

Polaris, though, may not be a candidate for GLXP; its goal of searching water ice at the pole is interesting. It is designed for drilling at moon's surface on polar region, which is characterised by low-glancing sun angles and near cryogenic temperatures. The rover is tall enough to deploy a 1.2m (4-feet) drill and produce 250W of power with solar panels oriented toward the sun. The rover will also prospect for water, oxygen, methane and other volatiles, which could be useful for energy, supporting life and producing rocket fuel. Astrobotic announced the completion of Polaris prototype on October 8, 2012.

Red Rover, a candidate for GLXP, is a mobile explorer that scouts the moon's mid-latitudes near Equator with cameras and science payloads. It navigates, detects obstacles and captures 3D video footage and maps.

The launch of Red Rover is planned by SpaceX Falcon 9 rocket. After achieving a low-Earth orbit, Falcon 9 stage two reignites for trans-lunar injection (TLI) to propel the spacecraft into a 4.5-day cruise to the moon.

Guidance, navigation and control software (developed at Carnegie Mellon, the USA) will keep the rocket on the right path. Due to lack of a global positioning system (GPS), the vehicle will be guided by plotting its trajectory to the moon by referring to stars, the moon and Earth. Once in moon's orbit, the spacecraft and rover must descend to the moon's surface.

It is interesting to note that Late Astronaut Neil Armstrong piloted the lunar module from orbit to the specific location on the moon, avoiding local hazards like boulders and craters. However, the Earth-to-moon distance imposes a time lag that does not facilitate real-time control from Earth, so the spacecraft's software must accomplish autonomously what Armstrong did manually.

A primary descent engine will burn to slow the spacecraft down as it approaches the moon, while small thrusters will keep the vehicle stabilised. Touching down two days after lunar dawn, Lander will deploy two ramps.

Bolts that hold the ramps folded against the ladder are rigged to break apart under immense heat. After the ramps fall from the spacecraft to the ground, the rover will roll down to the moon's surface. Binocular

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